

NAMPO, Korea, Democratic People's Republic of

WMO: 470600

Lat: 38.717N

Lon: 125.383E

Elev: 47

StdP: 100.76

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-11.0	-19.5	0.7	-11.4	-17.5	0.8	-9.6	10.9	-8.0	9.2	-5.5	3.0	320	0.401

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.1	30.9	26.2	29.6	25.2	28.5	24.4	27.0	30.0	26.2	28.7	25.4	27.6	1.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.6	29.0	25.4	20.7	28.0	24.8	19.9	27.1	85.5	30.2	81.8	28.9	78.7	27.7	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	7.6	6.3	DB	-15.0	32.7	2.8	1.4	-17.0	33.7	-18.7	34.5	-20.2	35.3	-22.3	36.3
			WB	-15.5	28.2	2.5	1.3	-17.3	29.1	-18.8	29.9	-20.2	30.6	-22.0	31.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-4.5	-1.9	4.1	11.1	17.2	21.8	25.3	20.8	14.3	5.8	-1.7	(6)	
(7)		DBStd	10.88	3.90	3.76	3.42	3.41	2.67	2.09	2.03	2.14	2.33	3.29	4.52	4.06	(7)
(8)		HDD10.0	1499	450	332	184	27	0	0	0	0	7	137	362	(8)	
(9)		HDD18.3	3120	708	566	440	217	55	2	0	7	129	375	621	(9)	
(10)		CDD10.0	2045	0	0	2	60	222	355	459	473	323	139	12	0	(10)
(11)		CDD18.3	624	0	0	0	1	18	107	201	214	80	3	0	0	(11)
(12)		CDH23.3	3743	0	0	0	3	89	440	1331	1610	264	7	0	0	(12)
(13)		CDH26.7	796	0	0	0	0	9	64	298	406	19	0	0	0	(13)
(14)	Wind	WSAvg	2.3	2.4	2.5	3.0	2.9	2.7	2.1	1.9	1.9	2.0	2.3	2.3	(14)	
(15)	Precipitation	PrecAvg	844	10	14	18	44	75	91	250	183	88	36	18	(15)	
(16)		PrecMax	1783	28	35	48	99	177	284	593	495	366	113	115	46	(16)
(17)		PrecMin	410	2	0	1	2	4	21	59	25	9	2	6	2	(17)
(18)		PrecStd	274	7	9	13	24	50	58	139	120	85	29	25	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	10.0	16.6	23.2	27.9	29.9	32.5	32.7	28.5	24.3	17.9	9.2	(19)
(20)		MCWB	4.1	6.0	11.7	15.7	19.3	21.9	26.7	27.6	23.0	18.4	13.9	7.3	(20)	
(21)		2%	DB	4.0	7.0	13.4	20.5	25.5	28.3	30.8	31.3	27.1	22.1	15.5	7.1	(21)
(22)		MCWB	2.1	4.4	8.9	14.4	17.9	21.9	26.1	26.8	22.1	17.3	12.9	5.2	(22)	
(23)		5%	DB	2.5	5.0	11.4	18.6	24.0	27.2	29.4	30.1	26.0	20.6	13.7	5.4	(23)
(24)		MCWB	0.9	2.6	7.8	13.0	17.1	21.2	25.4	26.0	21.1	16.3	11.4	3.7	(24)	
(25)		10%	DB	1.0	3.5	9.3	16.8	22.3	25.9	28.3	28.9	24.9	19.1	12.2	3.9	(25)
(26)		MCWB	-0.4	1.5	6.1	11.9	16.2	20.7	24.8	25.2	20.6	15.4	9.9	2.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	7.3	12.4	17.2	21.1	24.3	28.1	28.5	24.8	19.6	15.4	7.9	(27)
(28)		MDCB	5.5	8.6	15.5	21.5	26.2	27.7	31.1	31.6	26.9	22.7	16.9	9.1	(28)	
(29)		2%	WB	2.5	4.7	9.9	15.2	19.4	23.3	26.9	27.2	23.5	18.2	13.3	5.5	(29)
(30)		MDCB	3.7	6.6	12.6	19.4	23.7	26.7	29.7	30.4	25.7	21.0	14.9	6.8	(30)	
(31)		5%	WB	1.1	3.0	8.1	13.9	18.3	22.4	26.0	26.5	22.5	17.2	11.8	3.9	(31)
(32)		MDCB	2.3	4.8	10.6	17.6	22.1	25.5	28.5	29.4	24.8	19.7	13.6	5.2	(32)	
(33)		10%	WB	-0.2	1.7	6.5	12.5	17.2	21.6	25.3	25.9	21.5	16.2	10.2	2.4	(33)
(34)		MDCB	1.0	3.2	9.1	15.8	20.9	24.5	27.5	28.2	23.9	18.6	11.8	3.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	6.3	6.7	7.2	8.3	8.7	7.4	5.8	6.1	7.4	7.8	6.6	5.7	(35)	
(36)		5% DB	MCDBR	7.6	8.8	10.6	11.2	11.3	9.4	7.5	7.4	8.5	9.1	8.5	7.1	(36)
(37)		MCWBR	6.5	6.8	7.2	6.5	5.6	4.7	4.0	3.9	4.6	5.1	6.4	6.1	(37)	
(38)		5% WB	MCDBR	7.1	8.4	9.9	10.0	10.0	7.8	6.3	6.6	7.0	8.2	8.0	7.0	(38)
(39)		MCWBR	6.4	7.0	7.5	6.7	6.0	4.5	3.9	3.9	4.8	5.4	6.8	6.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.424	0.492	0.551	0.618	0.633	0.717	0.691	0.588	0.516	0.534	0.504	0.436	(40)	
(41)		taud	1.933	1.809	1.723	1.638	1.656	1.559	1.658	1.874	1.973	1.825	1.837	1.929	(41)	
(42)		Ebn at Noon	695	697	704	688	690	633	646	705	730	649	601	642	(42)	
(43)		Edh at Noon	137	178	215	248	248	274	246	194	165	173	151	129	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.39	3.19	4.38	5.12	5.69	5.23	4.19	4.49	4.38	3.52	2.35	2.05	(44)	
(45)		RadStd	0.20	0.24	0.34	0.37	0.52	0.53	0.54	0.53	0.37	0.22	0.24	0.14	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+87	+52

Nomenclature: See separate page